

Purdue University - PHYS 605

SYLLABUS

Pedagogical Methods for Physics Graduate Students

COURSE CREDIT HOURS: 2

INSTRUCTIONAL MODALITY: Face-to-Face

INSTRUCTOR:

N. Sanjay Rebello

Office: PHYS 232

E-mail: rebellos@purdue.edu

Office Hours: Wednesday 10:30-11:20 AM; By Appointment

GRADUATE TEACHING ASSISTANT:

Tess Hoover

Office: PHYS 196

E-mail: hooover92@purdue.edu

Office Hour: Thursday 2-3 pm

Required Text:

The Science of Learning Physics by Jose Mestre and Jennifer Docktor

Other reading materials will be provided in Brightspace

Meeting Room and Time:

PHYS 338 (or sometimes 331) – Tuesday 3:30-5:20 pm

In this course you will study and discuss recent developments in our understanding of how students learn. Emphasis will be placed on models of learning and teaching techniques that can be applied to the teaching of physics to university students. After completion of this course, you will have a better grasp of the issues and techniques pertaining to the teaching and learning of undergraduate science.

Course Goals:

- To introduce new students to their peers and share their teaching experience.
- To inform new students of available university/departamental teaching resources.
- To familiarize students with the obligations and responsibilities of being a graduate teaching assistant (GTA).
- To learn how to facilitate a recitation/lab class or lead a large classroom.
- To introduce students to elements associated with quality instruction.
- To provide an environment where students may practice their teaching skills and receive feedback on their performance.
- To inform students of some of the basic results from Physics Education Research.
- To promote reflection on the art of teaching.

Class Obligations:

This course depends upon your participation. The class will be discussion-based. While you may not be accustomed to this format, it is your responsibility to voice your opinions on the discussion topics. You are encouraged to speak, even if you disagree with your classmates' or instructor's opinion. If you are unsure about what is appropriate, talk to the instructor or the GTA.

1. **Attending Class** – Attending class on time is mandatory. Excessive tardiness will result in a loss of a portion of your daily participation points.
2. **Participation** – Actively Participating in Class and Discussion Boards - As stated above, you are expected to actively participate in discussions both face-to-face and online via *Brightspace*. Failure to participate in class discussions or activities will result in a deduction of participation points.
3. **Assigned Reading** – In order to keep the class from becoming a traditional lecture, you will be assigned readings so that the class can have a discussion of the assigned material. These readings will be made available on *Brightspace*. You are expected to read the assigned reading BEFORE coming to class and be ready to participate in a discussion about the reading. When you read the articles that are assigned, be reflective and think about how it pertains to your own teaching. Quizzes may be given if there appears to be an issue with students completing the reading assignments. You will also be expected to collaborate with a group of your peers.
4. **Micro-Teaching** – Twice during the semester you will prepare and present instructional material as you would in a classroom setting and in a virtual environment. You will be graded by the instructor and you will receive feedback from your peers. Micro-teaching and reflection on your micro-teaching will be assessed using a rubric provided in advance.
5. **Written Assignments and Discussion Boards** – There will be several written assignments throughout the semester. These will include summaries of assigned readings through "Reading Reflection" discussion boards and class discussions, Reflective Journaling boards, FERPA completion, as well as a teaching philosophy statement. The purpose is for you to synthesize what you have learned from the readings, classroom discussions, and reflections. All written assignments must be typed or entered into *Brightspace* discussion boards. No hand written assignments will be accepted. Additional directions for written assignments may be found on *Brightspace*. Late assignments will be accepted up to 24 hours after the due date with a 20% penalty. After this, no late work will be accepted without a legitimate excuse. All written assignments and journal posts should be written in a thoughtful manner with clear connections and reflective thought. Submitted assignments will be assessed using a rubric.

Grading Policy:

Grades in PHYS 605 are based on the following:

- 39% – Participation in in-class discussions
- 15% – Participation in discussion boards
- 21% – Written assignments (reflections, teaching philosophy, etc.)
- 25% – Micro-Teaching

Course Component	% of Total Grade	Total Points
Participation in Discussions – 13 Total (Class participation & Discussion Boards)	39%	390 pts (30 pts x 13)
Discussion Boards - 6 Total	15%	150 pts (25 pts x 6)

Written Assignments – 6 (<i>Self Reflection Journals (3), FERPA completion (1), Teaching philosophy (1), Resource Gathering (1)</i>)	21%	210 pts (35 pts x 6)
Micro-Teaching – 2 Total (<i>Lesson + Reflection</i>)	25%	250 pts (125 pts x 2)
Total For Course	100%	1000 pts

Point cut-offs:

A: 100 – 86%	D: 74 – 71%
B: 86 – 78%	F: 71 – 0%
C: 78 – 74%	

Incomplete

A grade of incomplete (I) will be given only in unusual circumstances. To receive an “I” grade, a written request must be submitted prior to December 16th of 2023 and approved by the instructor. The request must describe the circumstances, along with a proposed timeline for completing the course work. Submitting a request does not ensure that an incomplete grade will be granted. If granted, you will be required to fill out and sign an “Incomplete Contract” form that will be turned in with the course grades. Any requests made after the course is completed will not be considered for an incomplete grade.

Absences and Excused (EX) Grades:

Unexcused absences from meeting in class and Micro-Teaching will receive a score of zero. Written assignments must be submitted by the due dates; though extensions may be requested for illnesses and other documented absences. When an Excused (EX) grade is given for an approved absence, the grade assigned will be replaced by the average of your other scores in the relevant category at the end of the semester.

Excused grades will be given ONLY in these three circumstances:

- Serious Illness documented with a doctor’s note. Go to the Purdue University Student Health Center (PUSH). You should receive a slip from the Center with the date, doctor’s name and a telephone number, OR go to your private physician and obtain a written excuse. Take your excuse to your TA or instructor. (We only need a doctor’s excuse, not a diagnosis or any other personal information).
- Personal Crisis (e.g., automobile accident, death of a close relative) After contacting us, present the TA with written evidence of your excuse (e.g., police accident report, funeral announcement, ...).
- Required attendance at an official Purdue activity (e.g., exam conflict, athletic event, band concert, again supported by appropriate documentation). Go to room 144 of the Physics Building at least a week in advance of the event. The undergraduate office staff will give you further instructions.
- Weather conditions that make it impossible to get to the university.

In all cases, contact the instructor in advance (if at all possible) by email (preferably), in person, or by phone to make arrangements.

If you will be or have been away for five consecutive days, you must contact the Dean of Students office. They will send us a letter on your behalf.

Electronic Device

Please make sure to turn off and leave in your bag all cell phones, pagers, earpieces, earbuds, headphones, iPods, and all other electronic or noise-making devices before entering class. They are disruptive to your fellow classmates. Also, please refrain from bringing laptops to class unless instructed.

Attendance Policy

This course follows Purdue's academic regulations regarding attendance, which states that students are expected to be present for every meeting of the classes in which they are enrolled. Attendance will be taken at the beginning of each class and lateness will be noted. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to the instructor is not possible, the student should contact the instructor as soon as possible by email or phone. When the student is unable to make direct contact with the instructor and is unable to leave word with the instructor's department because of circumstances beyond the student's control, and in cases falling under excused absence regulations, the student or the student's representative should contact or go to the [Office of the Dean of Students website](#) to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted for cases of grief/bereavement, military service, jury duty, and parenting leave. For details, see the [Academic Regulations & Student Conduct section](#) of the University Catalog website.

If you must miss class at any point in time during the semester, please reach out to any of the Course Coordinators via email so that we can communicate about how you can maintain your academic progress. If you find yourself too sick to progress in the course, notify your adviser and notify me via email or Brightspace. We will make arrangements based on your particular situation.

Please coordinate with the instructor for any excused absences for meeting face-to-face. Otherwise, we do not anticipate conflicts for completing various course assignments online. However, in case of a conflict due to a University sponsored event, the student must inform the instructor at least one week prior to the conflict. Instructors shall not penalize a student who chooses to reschedule an assignment under these options.

Conflicts should be resolved between the course instructors and the student. Reasons for conflict should be taken under careful and reasonable consideration by the instructor and student. If conflict resolution is not possible, the examination shall take precedence, subject to appeal through the head of the department in which the course is offered.

Academic Integrity

[Purdue's Honor Pledge](#): *"As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."*

Any effort to represent somebody else's work as your own, or allowing your work to be represented as somebody else's, is cheating. Discussing your various writing assignments with your fellow students is

not cheating and is encouraged. However, when it comes time to write your assignment it is expected that you write it on your own, using your own words to express your ideas. **If a student is found cheating, he or she will receive an F for the course and be reported to the Dean of Students. In serious cases the Dean may suspend or expel the student from the University.**

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace table of contents, under University Policies.

Non Discrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies.

Diversity and Inclusion Statement

In our discussions, structured and unstructured, we will explore a variety of challenging issues, which can help us enhance our understanding of different experiences and perspectives. This can be challenging, but in overcoming these challenges we find the greatest rewards. While we will design guidelines as a group, everyone should remember the following points:

- We are all in the process of learning about others and their experiences. Please speak with me, anonymously if needed, if something has made you uncomfortable.
- Intention and impact are not always aligned, and we should respect the impact something may have on someone even if it was not the speaker's intention.
- We all come to the class with a variety of experiences and a range of expertise, we should respect these in others while critically examining them in ourselves.

Expectations Regarding Respect for Others

At Purdue, you will come into contact with many people who appear very different from you. Beyond superficial physical differences, members of our Purdue community may hold opinions with which you may disagree and may be very different from you along the infinite number of human dimensions. These physical, intellectual and cultural differences offer the opportunity to learn and enrich your experiences as a student and a human being. Our expectations of you as a member of Purdue and of the larger global community, is that you will at all times show respect for others. Learning how to do so is but one component of your personal growth and one way in which you, together with the rest of our community, can create a more inclusionary environment. It is your obligation as an individual with the privilege to attend an institution of higher learning to engage with others in respectful ways.

Purdue University is committed to providing a safe and secure campus environment for members of the university community. Purdue strives to create an educational environment for students and a work environment for employees that promote educational and career goals. Violent Behavior impedes such goals. Therefore, Violent Behavior is prohibited in or on any University Facility or while participating in

any university activity. See the Student Widget on our course Brightspace for more information on the Violent Behavior Policy.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try [WellTrack](#). Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the [Office of the Dean of Students](#). Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc. sign up for free one-on-one virtual or in-person sessions with a [Purdue Wellness Coach at RecWell](#). Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. If you have any questions, please contact Purdue Wellness at evans240@purdue.edu.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact [Counseling and Psychological Services \(CAPS\)](#) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Basic Needs Security

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday. Considering the significant disruptions caused by the current global crisis as it related to COVID-19, students may submit requests for emergency assistance from the [Critical Needs Fund](#)

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

Changes to Course Due to Major Campus Emergency

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Course information will, in general, be disseminated via one or more of the following routes: lecture, recitations, labs, email, and Brightspace.

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Students with disabilities must be registered with Disability Resource Center (DRC) (drc@purdue.edu) in the Office of the Dean of Students before classroom accommodations can be provided. If you are eligible for academic accommodations because you have a documented disability that will impact your work in this class, please schedule an appointment with me as soon as possible to discuss your needs.

Purdue University is committed to making learning experiences accessible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Course Evaluation Statement

During the last two weeks of the semester, you will be provided with an opportunity to give feedback on this course and your instructor. Purdue uses an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site. You will have up to 13 days to complete this evaluation. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

Copyright

Online educational environments, like all learning environments, should provide opportunities for students to reflect, explore new ideas, post opinions openly, and have the freedom to change those opinions over time. Students enrolled in and instructors working in online courses are the authors of the works they create in the learning environment. As authors, they own the copyright in their works subject only to the university's right to use those works for educational purposes (Visit [Purdue University Copyright Office](#)). Students may not copy, reproduce, or post to any other outlet (e.g., YouTube, Facebook, or other open media sources or websites) any work in which they are not the sole or joint author or have not obtained the permission of the author(s).

Changes to Syllabus and Schedule

The instructor reserves the right to make changes to this SYLLABUS and/or SCHEDULE for the course. Any changes will be posted on Brightspace and you will receive an email informing you of the changes.